

Ryan Lim

ARTIFICIAL INTELLIGENCE MAJOR

EDUCATION

(Dual Degree) Bachelor of Computer Science (Hons) (Artificial Intelligence) | 2023-2026

Asia Pacific University (APU) & De Montfort University

- Relevant Coursework: Python / Java / C++ / C# programming, Artificial Intelligence (Chatbot, Machine learning, NLP, Robotics, LLM), Backend Databases, Data Analysis, Imaging and Special effects in games, Data structures, Mobile App Engineering, Multi-threading, CUDA
- Results: **3.6 CGPA | FYP (96%)**

PROJECTS

AlzCare – AI Assisted Mobile Application for Alzheimer's Care (Flutter, FastAPI, Firebase)

Final Year Project | Machine Learning | Mobile App Development | Healthcare AI | SDG 3

- Designed and developed a full-stack mobile application supporting Alzheimer's patients and caregivers, combining medication adherence tracking, cognitive training games, and AI-driven personalization within a single system
- Engineered a Gated Recurrent Unit (GRU) neural network in PyTorch to analyze 14-day medication logs and predict non-adherence risk, enabling proactive caregiver alerts
- Implemented a Contextual Bandits reinforcement learning model using an Epsilon-Greedy policy to dynamically personalize cognitive game difficulty based on real-time patient performance
- Built a secure FastAPI backend integrated with Firebase Firestore and Authentication, handling role-based access control, real-time data sync, and REST API communication with the Flutter frontend
- Designed an accessibility-first UI/UX tailored for elderly users with cognitive impairments, featuring large icons, simplified navigation, and clear task-based layouts
- Conducted comprehensive testing (unit testing, black box testing, GRU model evaluation, and User Acceptance Testing with 5 target users) achieving 100% pass rate across 23 functional test cases
- Delivered exportable PDF health reports and a caregiver monitoring dashboard, directly reducing caregiver burden through remote, real-time patient oversight

E-Commerce Data Analyzer Using Data Structures in C++

Data Structures with C++

- Designed and implemented a C++ application to analyze real-world e-commerce transactions and customer reviews using arrays and linked lists
- Used 2-pass memory allocation strategy for dynamic arrays and real-time allocation for linked lists based on CSV file contents.
- Implemented and compared Selection Sort on both data structures to evaluate performance on sorting the transactions by date
- Applied a linear hash-style search to count the payment methods and combined hashing with sorting to find the most frequent words used in 1-star reviews.
- Highlighted trade-offs in time complexity ($O(n^2)$ for sort, $O(n)$ or $O(U)$ for hash lookups) and space efficiency between structures.

Dragon's Ring – 2D Kickboxing Game using Python

Game Development | Python | Pygame | Visual/Audio Effects

- Developed a fully playable 2D martial arts game using Python, inspired by classic arcade-style fighting mechanics and narrative-driven gameplay.
- Designed and implemented many themed stages (Underpass dojo, Wushi Dragon Grotto, etc.) with dynamic visuals, boss fights, and increasing complexity.
- Integrated cinematic effects like screen shaking, slow-motion, explosions, and animated transitions to improve the players immersion and feedback.
- Focused end-to-end development from concept and storyboarding to asset integration, audio syncing, and performance optimization

SavePlus+ – Personal Finance Tracker App (Flutter)

Mobile App Development | Flutter | Firebase | UI/UX Design | State Management

- Developed a cross-platform mobile application by using Flutter to help users manage their daily expenses, set some saving goals, and monitor budgeting trends
- Designed and implemented many features like user authentication using Firebase, expense/income tracking, savings goals, dark mode toggling, and financial reports visualized with graphs
- Integrated Firebase Authentication and Firestore for a more secure sign-in / sign-up and real time data storage across devices
- Applied state management techniques (ephemeral and app-level) to handle navigation states, theme preferences, and user session data more efficiently
- Conducted extensive manual and automated UI testing, including Firebase Test Lab Robo Tests, to ensure reliability and error handling in diverse usage scenarios.

UK Crime Data Analytics – Big Data & Machine Learning (Apache Spark)

Big Data Processing | Spark SQL | Machine Learning Pipelines | Cross-Validation | Data Visualization

- Processed and analysed large-scale crime datasets (2023–2024) from three UK counties using Apache Spark, combining monthly CSV files into a unified DataFrame and performing data cleaning, schema validation, and feature engineering.
- Implemented Spark SQL queries to compute county-wise and monthly crime statistics, identify highest/lowest periods, and rank crime types by frequency.
- Designed visual analytics — line, bar, bar charts, and pie charts — to reveal temporal and regional crime trends and present findings through descriptive summaries.
- Built and evaluated multiple machine learning models (e.g., Logistic Regression, Decision Tree, Random Forest) to predict crime categories and assess performance using accuracy metrics and k-fold cross-validation.
- Compared model results, optimized hyperparameters, and discussed implications of fairness and bias in predictive policing contexts.

Emergency Room Infection Simulation – Agent-Based Modelling (NetLogo)

Parallel Computing | Multi-Threading | Agent-Based Simulation | Performance Optimization

- Developed a multi-agent emergency room simulation to understand infection spread in times with an infection/virus spreading with implementation of multi-threading with multiple agents working simultaneously.
- Designed agent behaviors for patients, nurses, treatment rooms, and exit door, by incorporating realistic arrival rates, priority based queues, treatment durations, and infection transmission mechanisms
- Implemented parallel processing to handle simultaneous agent updates, including queue movement, patient state, treatment processing, and room states, enabling a scalable simulation.
- Conducted multiple real-life scenarios (baseline settings, masking policies, overcrowding, and high transmission environments) to successfully determine causes of infection spread under separate conditions.
- Applied CUDA fundamentals in tutorial exercises, implementing GRU-based parallel kernels to explore thread execution, memory management, and performance acceleration concepts.

WORK EXPERIENCE

Website Manager (Internship) | June 2024 - November 2024

The Knowledge Group (m) SDN BHD, Kuala Lumpur

- Handled Data entry for client listings
- Update Company's website on changes in products like pricing, product image, product name, etc.
- broadcast promotion leaflets

CERTIFICATES

AI & Large Language Models (LLMs) Learning Program | November 2025

Coursera / OpenAI Learning Module

- Gained hands-on understanding of LLM architecture, training, and fine-tuning methods
- Explored ethical, environmental, and data privacy considerations in AI deployment
- Studied multimodal vs. non-multimodal model capabilities and real-world applications
- Applied concepts of text preprocessing, representation, and RLHF (Reinforcement Learning with Human Feedback)

Vibe Coding 101 | July 2026

PDX2026 / Kracked Devs

- Completed a workshop on building AI-enabled web experiences
- Explored practical front-end and product-oriented development ideas
- Strengthened understanding of rapid prototyping and implementation workflows

AI Agents: From Zero to Production Workshop 1 | 2 July 2026

APU Hackthletes

- Participated in a hands-on workshop focused on AI agents and production-oriented thinking
- Learned practical ideas for moving agent-based systems from prototype concepts toward usable applications
- Strengthened exposure to current applied AI development practices